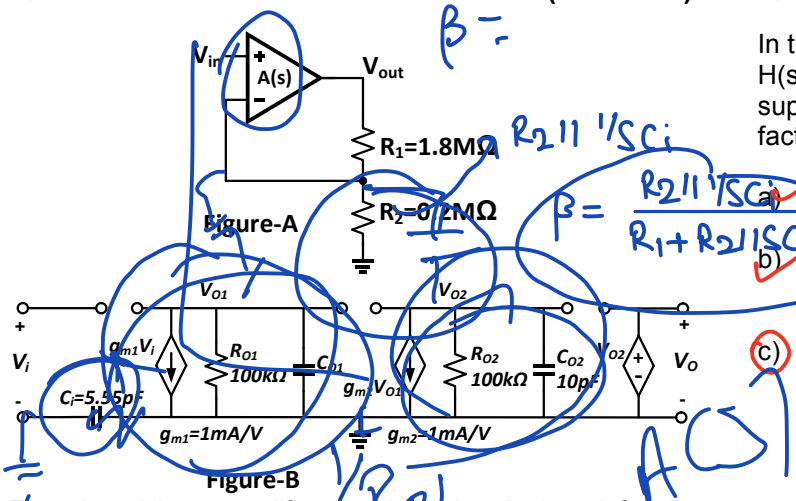
Quiz-II (Jan.-May 2021)

EE2019 – Analog Systems and Lab

Total Marks: 30 **Duration: 45 min**

Question-1 (15 marks)



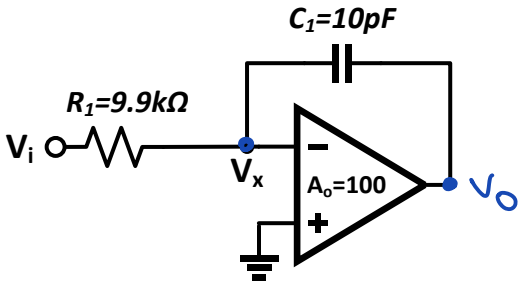
The closed loop amplifier needs to be designed for the stable operation as shown in Figure-A.

a) Find the loop gain transfer function, poles, and zeroes. *(5 marks)*

b) Calculate the value of capacitor C_{O1} to achieve the phase margin of 60 degrees. *(5 marks)*

c) Draw the Bode magnitude and phase plots of the loop gain transfer function. Clearly mention values of gain, phase, location of poles, zeroes and unity gain frequency. *(5 marks)*

Question-2 (6 marks)



For the circuit shown above:

a) Find the transfer function $V_x(s) / V_i(s)$ in terms of R_1 and C_1. *(2 marks)*

b) Find the expression for $V_x(t)$ *for unit step input* and draw $V_x(t)$ waveform. *(2 marks)*

c) Find the value of ***RC*** time constant in the above settling. *(2 marks)*

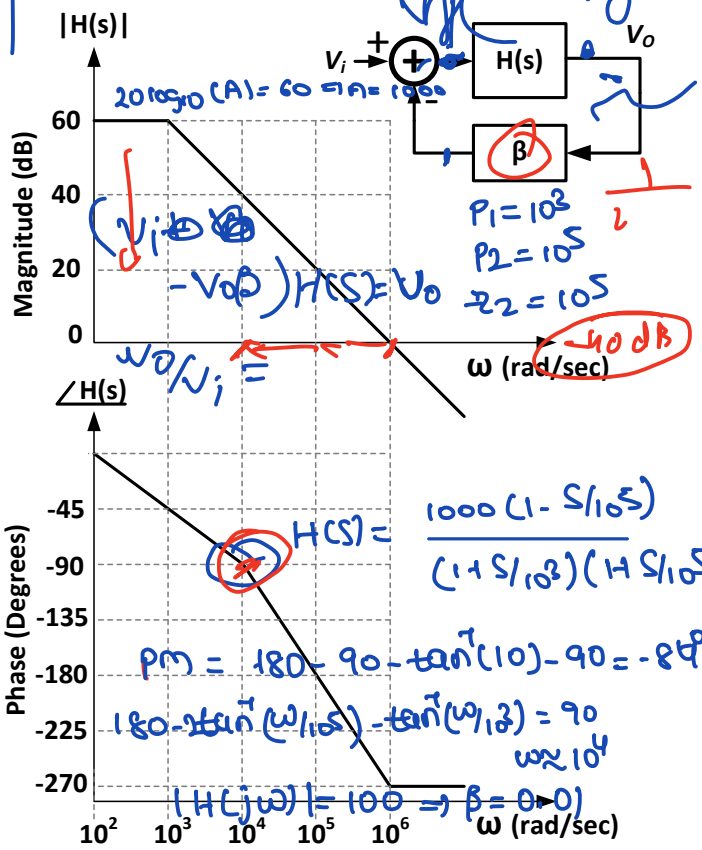
Question-3 (9 marks)

In the figure shown below, a system with transfer function, H(s) of the given Bode magnitude and phase response, is supposed to be operated in the closed loop with feedback factor, β.

a) Find the transfer function H(s). *(3 marks)*

b) Determine if the system is stable or unstable for β=1. *(3 marks)*

c) Find the value of β to achieve the phase margin of 90 degrees. *(3 marks)*

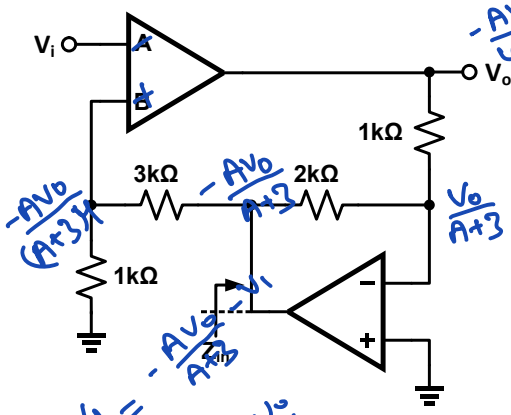


Quiz-II (Jan.-May 2021)

EE2019 – Analog Systems and Lab

Total Marks: 30 **Duration: 45 min**

Question-1 **(10 marks)**



Assume that the each opamp has input-to-output gain as $A(s) = 10^6/s$.

a) Find the signs of terminals A and B for the above circuit to operate in dc negative feedback. *(2 marks)*

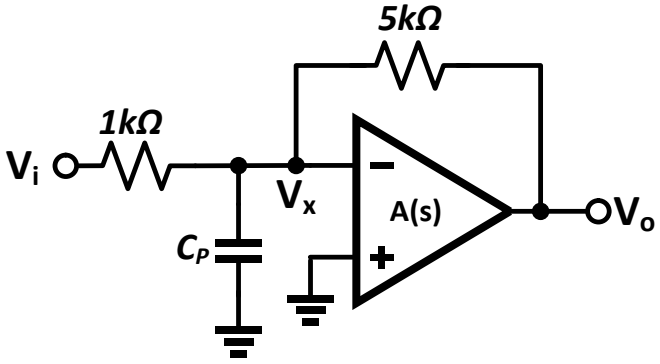
b) Find the transfer function $V_o(s)/V_i(s)$. *(4 marks)*

c) Find the input impedance $Z_{in}(s)$ in the above circuit. *(4 marks)*

d) What is the value of C_p for which the circuit becomes unstable? *(5 marks)*

e) With C_p being the value calculated in the previous part, can you change the circuit so that the phase margin is 60° without changing the opamp or the closed-loop dc gain V_o/V_i? *(5 marks)*

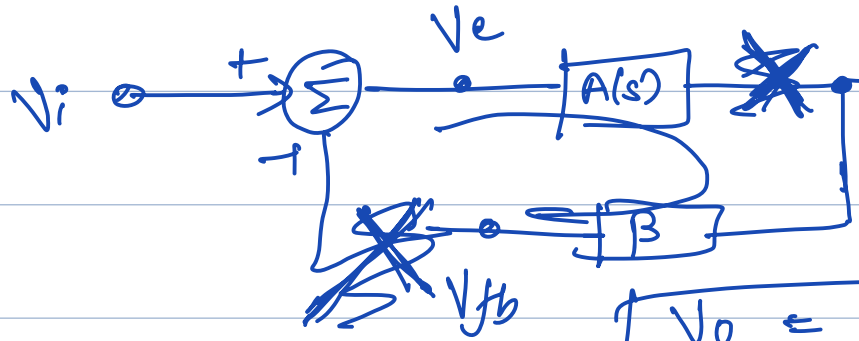
Question-2 **(20 marks)**



Above figure shows an inverting amplifier. The opamp has a gain $A(s) = A_0/(1+s/p_1)(1+s/p_2)$ where $A_0 = 20000$, $p_1 = 1$ krad/s, $p_2 = 10$ Mrad/s. C_p is a parasitic capacitor.

a) What is the phase margin of the system with $C_p = 0$? *(5 marks)*

b) What is the closed loop bandwidth of the system? *(5 marks)*



CLTF
CLG

$$\frac{V_o}{V_i} = \frac{A\beta}{1+A\beta} \cdot \frac{1}{\beta}$$

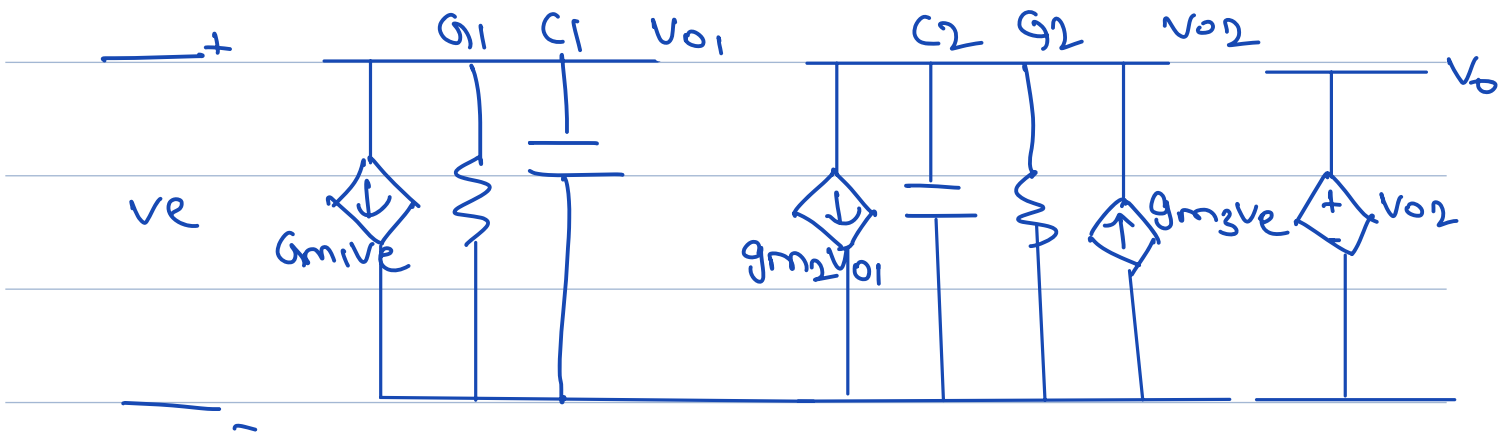
$$\frac{V_e}{V_i}(s)$$

$$\frac{V_o}{V_e}(s) = A(s)$$

$$\frac{V_{fb}}{V_o} = \beta$$

$$L(s) = A(s)\beta$$

$$L(s) = A(s)\beta$$



$$V_{o1}G_1 + V_{o1}(sC_1) + G_{m1}V_e = 0 \Rightarrow \frac{V_{o1}}{V_e} = \frac{-G_{m1}}{G_1 + sC_1}$$

a) if $G_{m3} = 0 \Rightarrow \frac{V_o}{V_e} = \frac{G_{m1}G_{m2}}{(G_1 + sC_1)(G_2 + sC_2)}$

poles: G_1/C_1, G_2/C_2 zeros: nil

b) $V_{o2}(G_2 + sC_2) + g_{m2}\left(\frac{-g_{m1}}{G_1 + sC_1}\right)V_e = g_{m3}V_e$

$$\Rightarrow \frac{V_{o2}}{V_e} = \frac{g_{m3} + g_{m1}g_{m2}/G_1 + sC_1}{(G_2 + sC_2)}$$

c) get zero $= p_2 \Rightarrow \frac{G_{m1}G_{m2}}{G_{m3}C_1} = \frac{G_2}{C}$